

BIS REFERENCE HANDBOOK: HVAC, REFRIGERATION & RENEWABLE ENERGY

STANDARDS COMPENDIUM FOR AIR CONDITIONERS, SOLAR PV & GREEN TECH

Governed by PCD, ETD, and MED committees to ensure energy efficiency, thermal comfort, and renewable energy adoption.

1. ROOM AIR CONDITIONERS (IS 1391 PART 1 & 2 / IS/IEC 60335-2-40)

- Window and Split Air Conditioners testing for Cooling Capacity, ISEER (Indian Seasonal Energy Efficiency Ratio), and safety under low-GWP flammable refrigerants (R-32, R-290).

2. SOLAR PHOTOVOLTAIC MODULES & INVERTERS

- **IS 14286 / IEC 61215:** Terrestrial photovoltaic (PV) modules design qualification and type approval (damp heat test, thermal cycling, mechanical load test 5400 Pa).
- **IS/IEC 61730 (Part 1 & 2):** PV module safety qualification (fire test, electrical shock hazard).
- **IS 16221 (Part 2) & IS 16169:** Solar utility grid-tied inverters safety and anti-islanding protection.